This week

- Today: EDA examples

- Lab 3 and HW3 due tonight
  - Videos of my going through the assignments will be posted tomorrow
  - For figuring out what you missed and why, and reference for the midterm

- P2 data and description, lab 4 and HW4 due next Tuesday
  - Videos for lab 4 and HW4 will be posted immediately that night

This week

- Next week: Review and Unsupervised Learning
  - Tuesday: Clustering in first half of class, review in second half
    - Come with questions about content, assignments etc…
  - Thursday: PCA

- Midterm assigned after class Tuesday
  - No collaboration
  - Will cover all course content thru the end of week 4.
    - Includes text and regex

Exploration

Three objectives

1. **Exploration**: the process of learning about your dataset

2. **Inference**: using your data to learn about the world

3. **Prediction**: using your observed data to predict unobserved data

Most of what you will do as a data scientist will be in the service of one of these objectives

Exploration

Data **exploration** is the process of learning about your dataset

- Describe variables
  - Data type, distribution, range and outliers, etc…
- Uncover associations
  - Differences by category, linear or nonlinear relationships, etc…
- Identify and rectify problems
  - NAs, incorrect data types, erroneous values, etc…

Exploration

Data **exploration** is the process of learning about your dataset

It is an open-ended, creative process

→ No obvious metrics of success

- Can be the primary objective: What are the characteristics of our sample?
- More typically, the first step leading up to Prediction and Inference

Exploration

Describe your variables

```
pd.crosstab(penguins["species"], penguins["island"])
```

island	Biscoe	Dream	Torgersen
species			
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

```
penguins.describe()
```

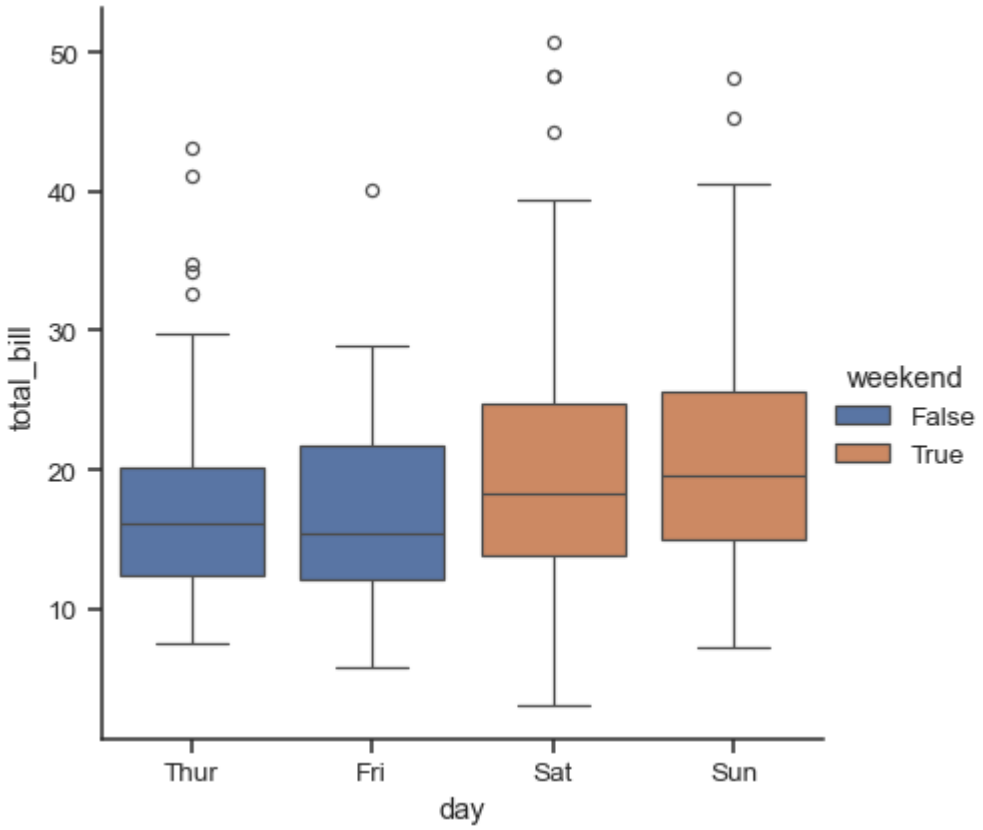
	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	year
count	342.000000	342.000000	342.000000	342.000000	344.000000
mean	43.921930	17.151170	200.915205	4201.754386	2008.029070
std	5.459584	1.974793	14.061714	801.954536	0.818356
min	32.100000	13.100000	172.000000	2700.000000	2007.000000

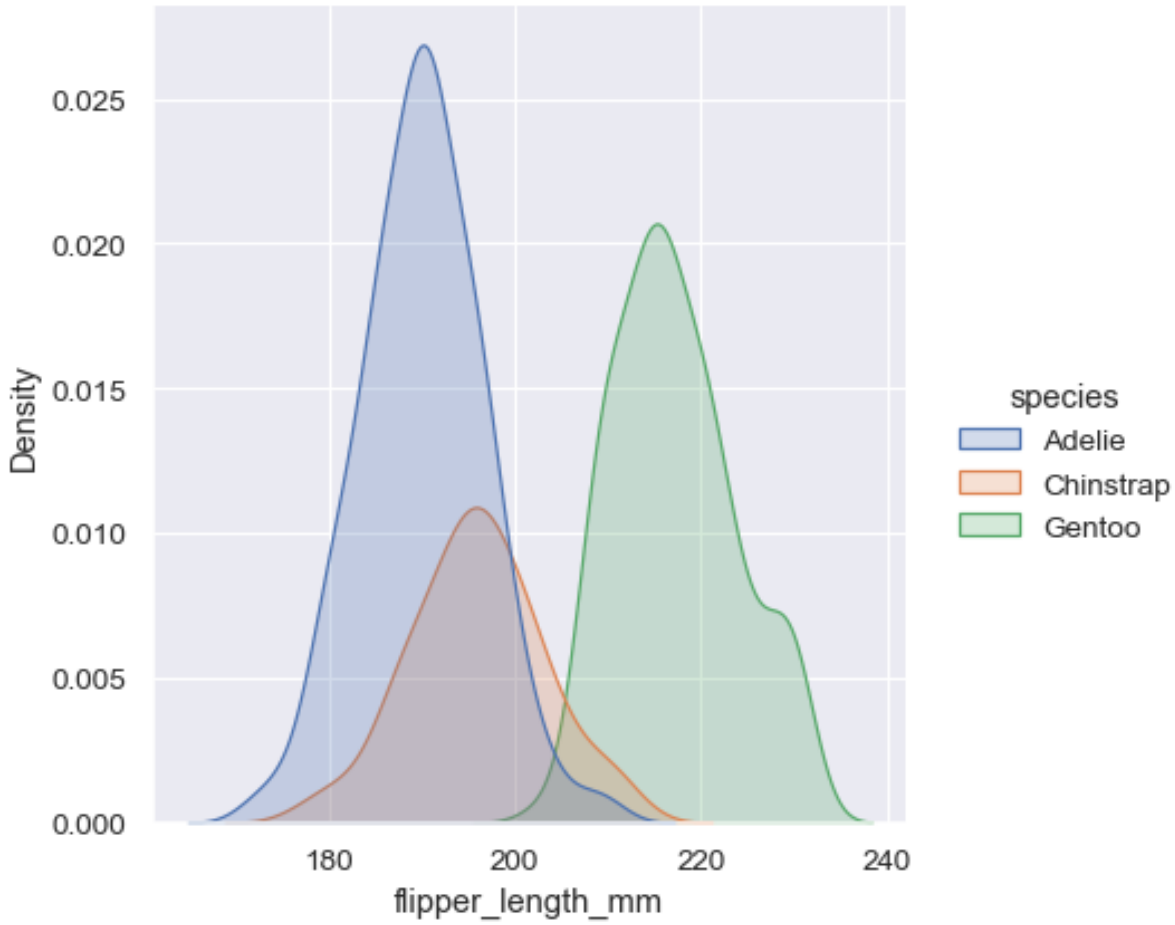
```
mottos["Language"].unique()
```

```
array(['Latin', 'English', 'Samoan', 'Greek', nan, 'Hawaiian', 'Italian',
       'French', ' Latin', 'Spanish', 'Chinook Jargon'], dtype=object)
```

Exploration

Visualize your data



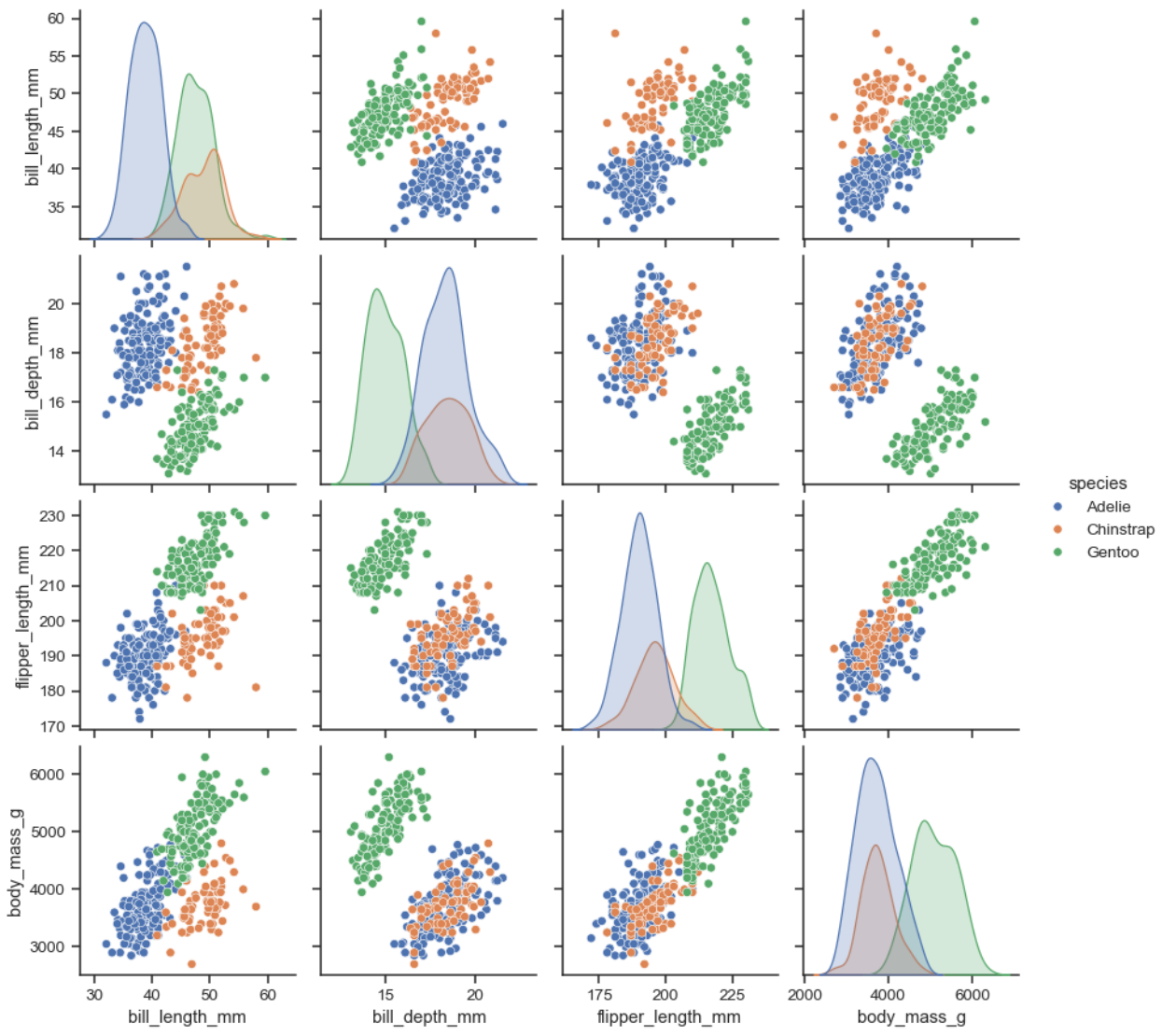


Exploration

Explore associations

- Visualize quantitative relationships
- Quantify correlations

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
bill_length_mm	1.000000	-0.235053	0.656181	0.595110
bill_depth_mm	-0.235053	1.000000	-0.583851	-0.471916
flipper_length_mm	0.656181	-0.583851	1.000000	0.871202
body_mass_g	0.595110	-0.471916	0.871202	1.000000



Exploration

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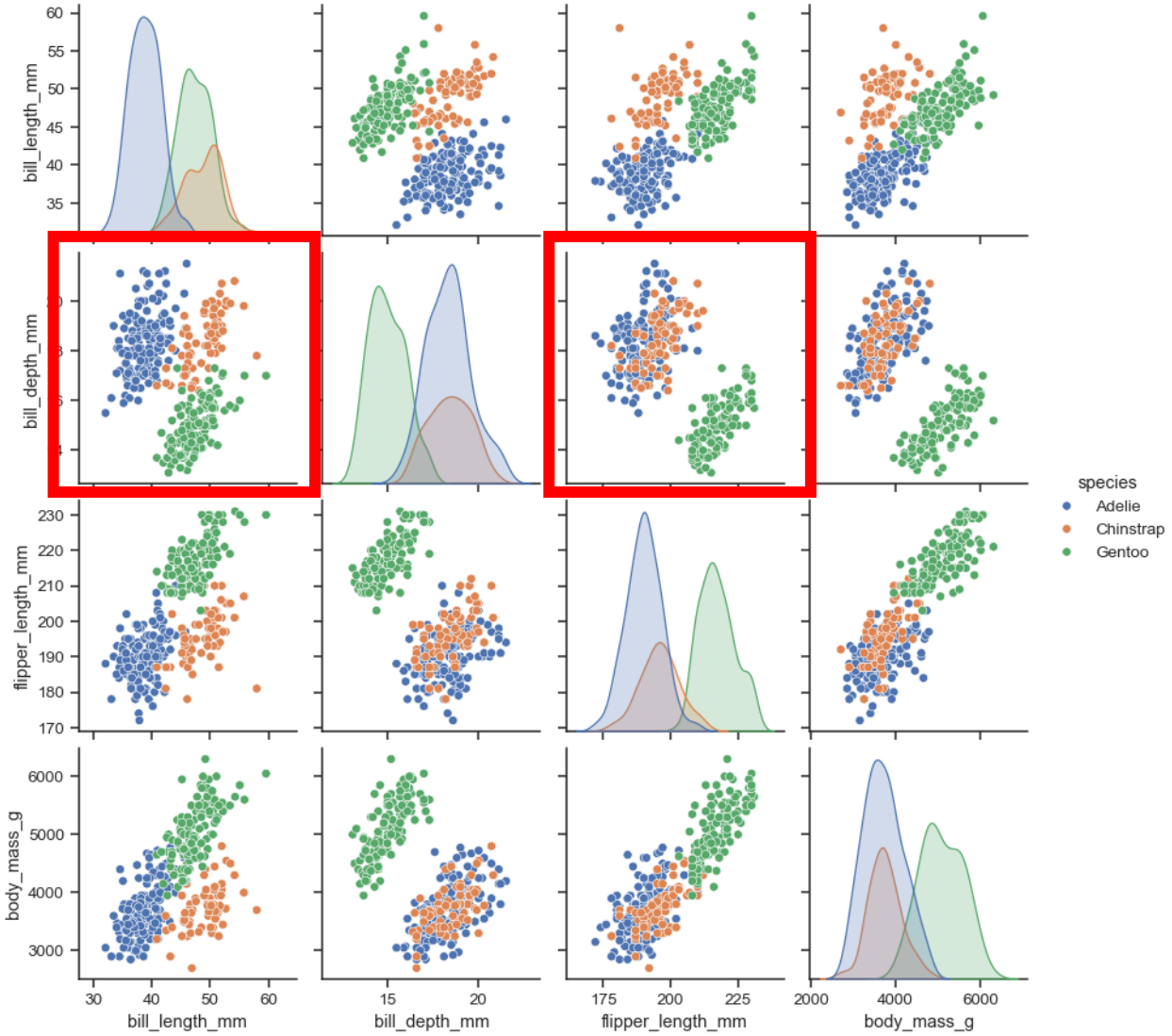

Exploration

Simpson's Paradox!

Explore associations

- Visualize quantitative relationships
- Quantify correlations

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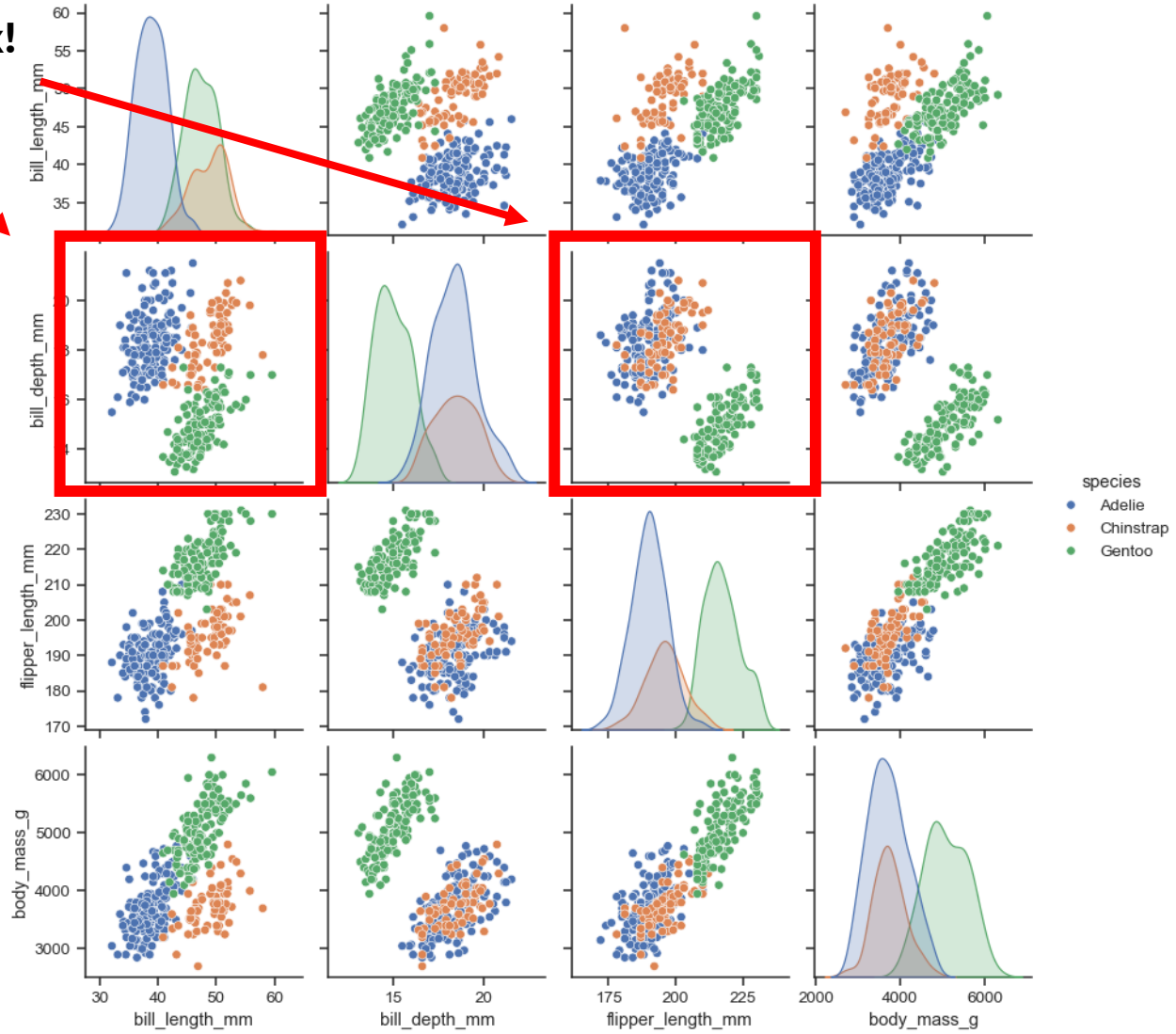

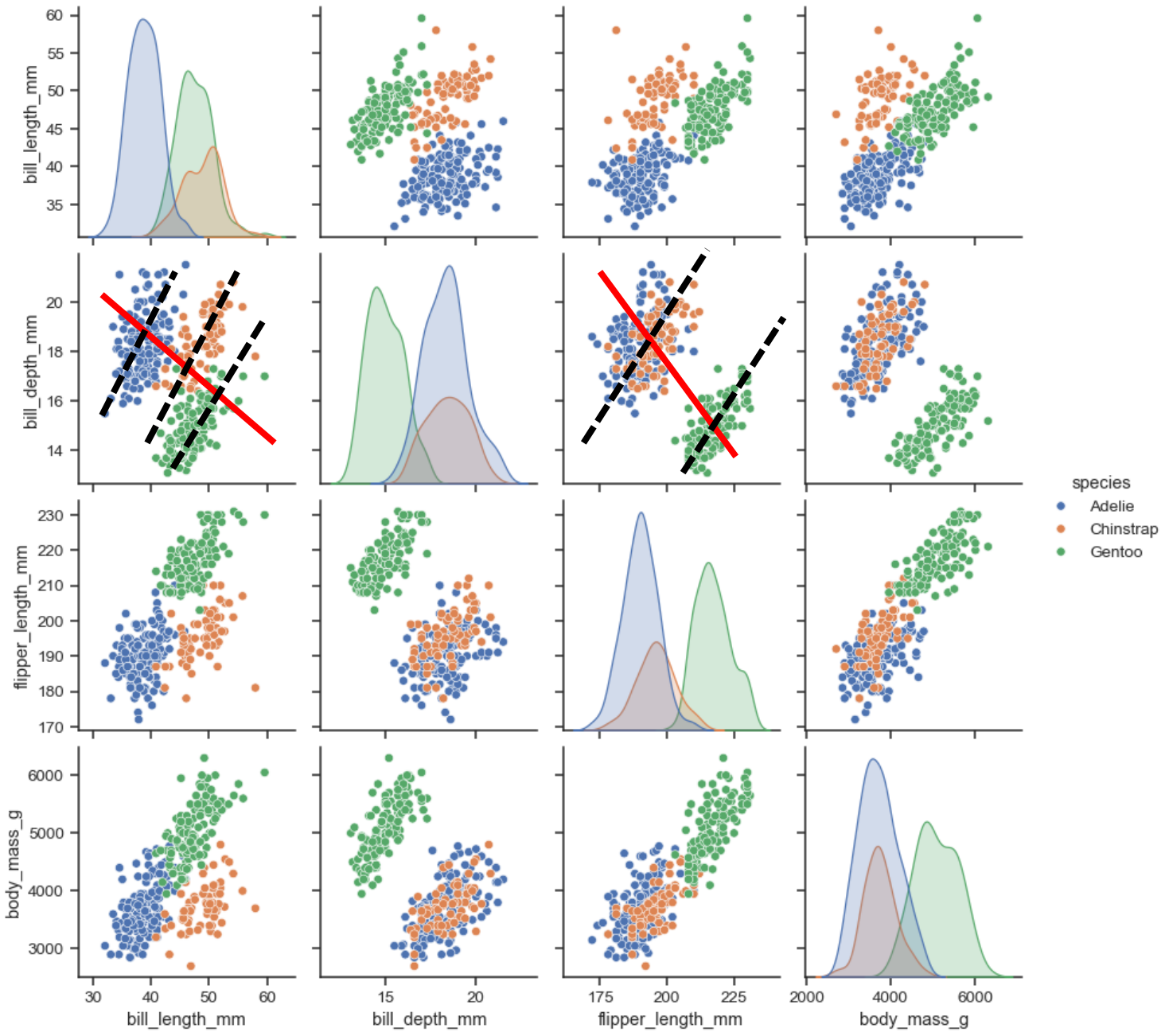
bill_length_mm
bill_depth_mm
flipper_length_mm
body_mass_g
species
Adelie
Chinstrap
Gentoo

Exploration

Clean your data

Identify problems that make analysis difficult or misleading

```
life.columns

Index(['Country', 'Year', 'Status', 'Life expectancy ', 'Adult Mortality',
       'infant deaths', 'Alcohol', 'percentage expenditure', 'Hepatitis B',
       'Measles ', ' BMI ', 'under-five deaths ', 'Polio', 'Total expenditure',
       'Diphtheria ', ' HIV/AIDS', 'GDP', 'Population',
       ' thinness  1-19 years', ' thinness 5-9 years',
       'Income composition of resources', 'Schooling'],
      dtype='object')
```

Exploration

```
life.columns
```

```
Index(['Country', 'Year', 'Status', 'Life expectancy ', 'Adult Mortality',
       'infant deaths', 'Alcohol', 'percentage expenditure', 'Hepatitis B',
       'Measles ', ' BMI ', 'under-five deaths ', 'Polio', 'Total expenditure',
       'Diphtheria ', ' HIV/AIDS', 'GDP', 'Population',
       ' thinness  1-19 years', ' thinness 5-9 years',
       'Income composition of resources', 'Schooling'],
      dtype='object')
```

```
life.columns.str.strip()
```

```
Index(['Country', 'Year', 'Status', 'Life expectancy', 'Adult Mortality',
       'infant deaths', 'Alcohol', 'percentage expenditure', 'Hepatitis B',
       'Measles', 'BMI', 'under-five deaths', 'Polio', 'Total expenditure',
       'Diphtheria', 'HIV/AIDS', 'GDP', 'Population', 'thinness  1-19 years',
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